



People Analytics

Turnover insights and risk drivers

This presentation summarizes an analysis of employee turnover to identify drivers of attrition and actionable retention strategies.

The study combines *Python*, *SQL*, and Power BI to deliver an interactive dashboard and data-driven recommendations.





Objective

This project was developed to analyze employee turnover patterns and identify the key factors associated with employee attrition.

Business Questions

- What is the overall turnover rate?
- Which departments experience the highest attrition?
- Do lower-paid employees leave more often?
- Does tenure influence employee turnover?
- Does job satisfaction impact retention?

Tools Used

- Python (Pandas, SQL, Seaborn)
- Power BI
- SQL Queries
- Data Visualization



Data Preparation & Methodology

Data Preparation

- Cleaned and standardized HR data.
- Created a numerical attrition indicator (Yes = 1, No = 0).
- Exported the processed dataset for Power BI reporting.

Analytical Approach

Using SQL and Python, I analyzed:

- Turnover rate by department
- Salary versus attrition
- Job satisfaction versus attrition
- Overtime versus attrition
- Employee risk profiles

Dashboard Development

The final dataset was integrated into Power BI to create an interactive HR analytics dashboard.



Key Findings

Overall Attrition

The company turnover rate is approximately 16%.

Department Analysis

Sales showed the highest turnover rate, followed by Human Resources.

Employee Satisfaction

Employees with lower job satisfaction were significantly more likely to leave the company.

Overtime Impact

Employees working overtime demonstrated substantially higher attrition rates than those without overtime.



Business Impact & Conclusion

Business Value

This analysis enables HR teams to:

- Identify departments with higher turnover risk.
- Monitor employee satisfaction levels.
- Reduce attrition-related recruitment and training costs.
- Develop targeted retention strategies.
- Support data-driven workforce planning.

Conclusion

The project demonstrates how HR data can be transformed into actionable business insights.

The strongest drivers of attrition identified were:

- Low job satisfaction
- Overtime work
- Lower compensation levels
- Department-specific workforce challenges

By combining Python, SQL, and Power BI, the solution provides a practical framework for proactive employee retention management.



Risk Factors

Primary drivers: *low job satisfaction, frequent overtime, and below-average salary.*

Secondary contributors include short tenure in certain roles and department-specific workload imbalances. Combine factors raise individual risk substantially.



Business Impact

Targeted retention can reduce recruiting and training costs and stabilize critical teams.

Use **risk segmentation** to prioritize interventions, measure satisfaction improvements, and track attrition trends via the Power BI dashboard.



Conclusions

Focus retention efforts on high-risk cohorts identified by satisfaction, overtime, and compensation gaps.

Implement targeted programs, monitor via the dashboard, and iterate with data-driven interventions to lower turnover and maintain workforce continuity.